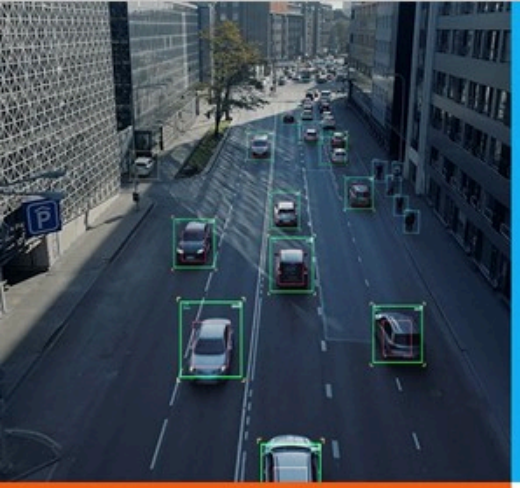

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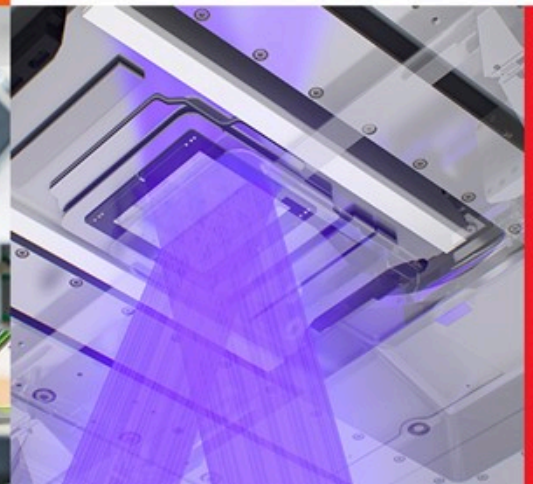

SOFTWARE

# C++ Training

## Strong types

# Strong types

```
void create(double mass, double length);
double mass;
double length;
create(mass, length);
create(length, mass);
```

- Typically a developer is not interested if an object stores its mass as double, float or int.
- We assume the library works and just use the API.

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Type aliases are not enforced by compiler, only syntactic. But an improvement nonetheless.

```
using Mass=double;
using Length =double;
void create(Mass mass, Length length);
Mass mass;
Length length;
create(mass, length);
create(length, length);
```

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- A STRONG type is a type used in place of another type to carry specific meaning through its name.
- Not an alias or typedef

```
struct Mass{};
struct Length{};
void create(Mass mass, Length length);
```

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```
template <typename T, typename Tag>
class StrongType
{
public:
    explicit StrongType(T v):_v(std::move(v)){}
private:
    T _v;
};
using Mass = StrongType<double, struct MassTag>;
using Length = StrongType<double, struct LengthTag>;
```

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- Conversion between units of the same tag can be implemented using std::ratio.
- std::chrono library to support custom ratios as a template parameter.
- There's a compiler in C++, use it as much as you can.
- You can create custom ratios but the SI ratios are provided by the standard library, for example std::kilo

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```
template <typename T, typename Tag, typename Ratio>
class StrongType
{
public:
    explicit StrongType(T v):_v(std::move(v)){}

    template <typename OtherRatio>
    operator StrongType<T, Tag, OtherRatio>() const
    {
        return StrongType<T, Tag, OtherRatio>(_v * Ratio::num / Ratio::den
            * OtherRatio::den / OtherRatio::num);
    }
    friend std::ostream& operator<<(std::ostream& os, StrongType const& x)
    {
        os << x._v;
        return os;
    }
private:
    T _v;
};
```

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## Curiously recurring template pattern (CRTP)

- For some types it could make sense to add/substract/multiply them, while for others it may not.
- Some applications may have no use for multiplied masses -$kg^2$ - in which case, multiplication should be blocked.

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## Slides simplified by removing ratio

```
template <typename StrongType>
struct Crtp
{
    StrongType const& Underlying() const { return static_cast<StrongType const&>(*this);}
};
template <typename StrongType>
struct Addable: Crtp<StrongType>
{
    StrongType operator+(StrongType const& other) const
    {
        return StrongType(this->Underlying().get() + other.get());
    }
};
```

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```
template <typename T, typename Tag, template<typename> class... Skills>
class StrongType: public Skills<StrongType<T, Tag, Skills...>>...
{
public:
    explicit StrongType(T v):_v(std::move(v)){}

    T get() const{return _v;}
    friend std::ostream& operator<<(std::ostream& os, StrongType x)
    {
        os << x._v;
        return os;
    }
private:
    T _v;
};
using Mass= StrongType<double, struct MassTag, Addable>;
```

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CRTP: Pass derived class as template parameter to base class

```
#include "strongType.h"

int main()
{
    Mass x(10);
    Mass y(4);

    std::cout << x+y<<std::endl;
}
```

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## Create look-up table to print ratios (not included in lib)

```
template<typename R>
constexpr std::string_view get_prefix() {
    if constexpr (std::is_same_v<R, std::kilo>) return "k";
    if constexpr (std::is_same_v<R, std::milli>) return "m";
    if constexpr (std::is_same_v<R, std::mega>) return "M";
    if constexpr (std::is_same_v<R, std::ratio<1>>) return "";
    return "?"; // Unknown scale
}
template <typename T, typename Tag, typename Ratio>
struct std::formatter<StrongType<T, Tag, Ratio>> : std::formatter<T> {
    auto format(const StrongType<T, Tag, Ratio>& st, format_context& ctx) const {
        auto it = std::formatter<T>::format(st.raw(), ctx); // returns iterator into buffer
        return std::format_to(it, " {}{}", get_prefix<Ratio>(), Tag::unit); //start writing at iterator
    }
};
```

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```
int main()
{
    using Meters = StrongType<double, DistanceTag>;
    using KiloMeters = StrongType<double, DistanceTag, std::kilo>;

    KiloMeters roadTrip{1.5};
    Meters inMeters = static_cast<Meters>(roadTrip); // Explicit for clarity

    std::cout << std::format("Distance A: {}\n", roadTrip);
    std::cout << std::format("Distance B: {}\n", inMeters);
}
```

```
Distance A: 1.5 km
Distance B: 1500 m
```

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# Zero-Cost: Standard Layout & Triviality

```
std::is_standard_layout_v<T>
```

- no virtual functions
- no virtual base classes
- non-static members have the same access control.
  "C-compatible."

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```
std::is_trivially_copyable_v<T>
```

- can be copied by simply copying its raw bits (like memcpy).
- No complex copy/move-constructors are called.
- CPU can move your type as a single block of data.

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- We want the safety at compile time
- No runtime overhead
- Assert at compile time, a double is still a double.
- Doubling down on, so to speak

```
static_assert(sizeof(Meters) == sizeof(Meters::value_type), "inheritence/member var added?");
static_assert(std::is_standard_layout_v<Meters>, "vtable in there?");
static_assert(std::is_trivially_copyable_v<Meters>, "copy/move ctor was added?");
```

Checks EBO of tags.

Avoid using references

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Do not reinvent the wheel.

boost::units is BIS when working with units.

```
using namespace boost::units;
using namespace boost::units::si;
quantity<energy> work(const quantity<force>& F, const quantity<length>& dx)
{
    return F*dx; // Defines the relation: work = force * distance.
}
int main()
{
    quantity<force> F(2.0 * newton);
    quantity<length> dx(2.0 *meter);
    quantity<energy> E(work(F,dx));
    std::cout<<"F  = " << F << std::endl <<"dx = " << dx << std::endl <<"E  = " << E << std::endl <<std::endl;
    return 0;
}
---
F  = 2 N
dx = 2 m
E  = 4 J
```

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```
using namespace boost::units;
using namespace boost::units::si;
quantity<length> work(const quantity<force>& F, const quantity<length>& dx)
{
    return F*dx;
}
---
<source>:15:13: error: could not convert 'boost::units::operator*(const quantity<Unit1, X>&, const
quantity<Unit2, Y>&) [with Unit1 = unit<list<dim<length_base_dimension, static_rational<1> >,
list<dim<mass_base_dimension, static_rational<1> >, list<dim<time_base_dimension, static_rational<-2> >,
dimensionless_type> > >, homogeneous_system<list<si::meter_base_unit,
list<scaled_base_unit<cgs::gram_base_unit, scale<10, static_rational<3> > >, list<si::second_base_unit,
list<si::ampere_base_unit, list<si::kelvin_base_unit, list<si::mole_base_unit,
list<si::candela_base_unit, list<angle::radian_base_unit, list<angle::steradian_base_unit,
dimensionless_type> > > > > > > > > > >; Unit2 = unit<list<dim<length_base_dimension, static_rational<1>
>, dimensionless_type>, homogeneous_system<list<si::meter_base_unit,
list<scaled_base_unit<cgs::gram_base_unit, scale<10, static_rational<3> >
```

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## exercises

strong_types/ex3.cpp

strong_types/ex4.cpp

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